

SplitStack Functional Design Diagrams

Version: 1.0

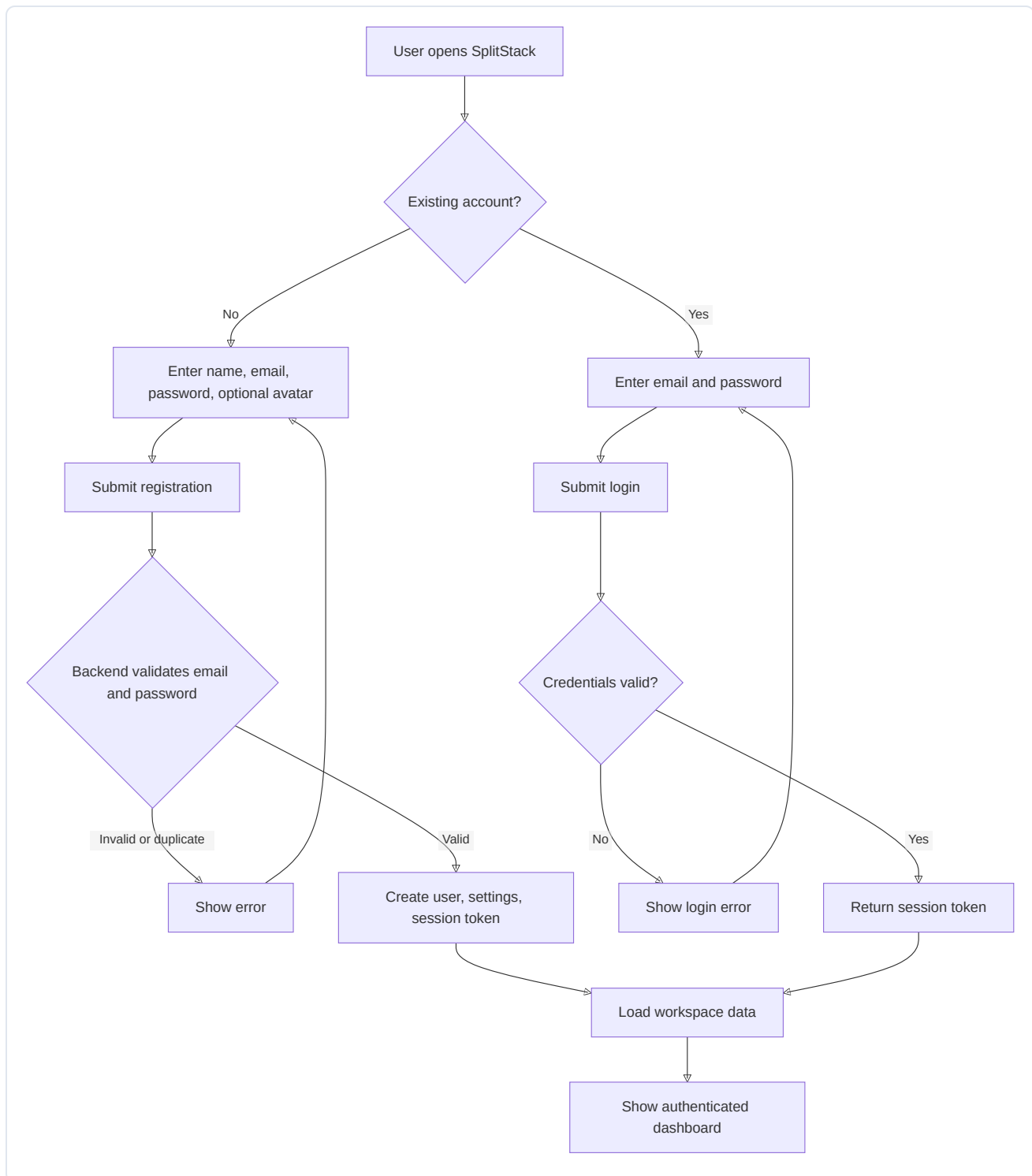
Date: May 12, 2026

Related documents: `planning files/Project_Requirements_Specification (2).docx`, `planning files/SplitStack_Use_Cases (1) (3).docx`

Purpose

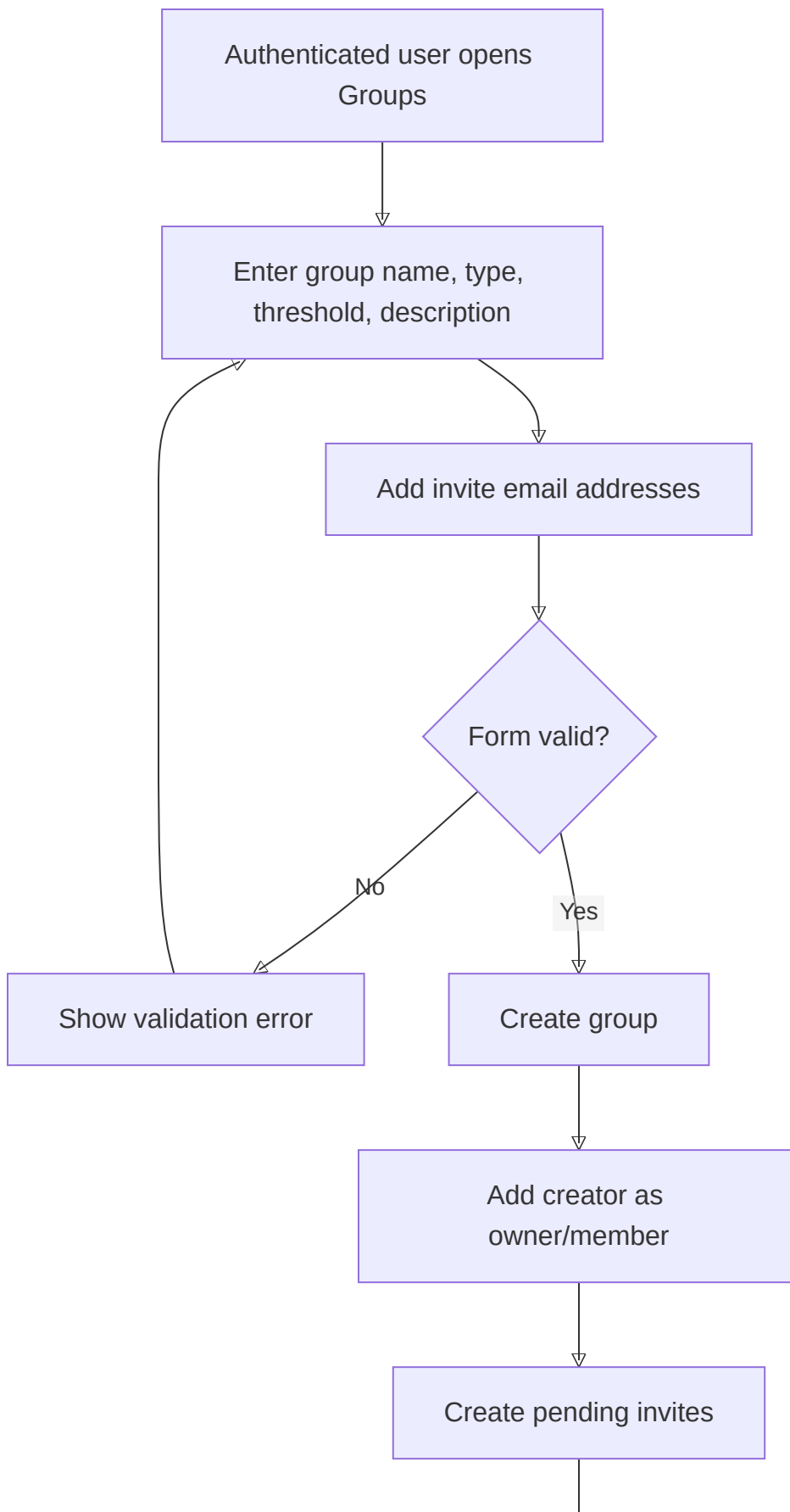
This document supplements the functional requirements and use-case documents with activity diagrams and a domain/ER model for the implemented SplitStack system.

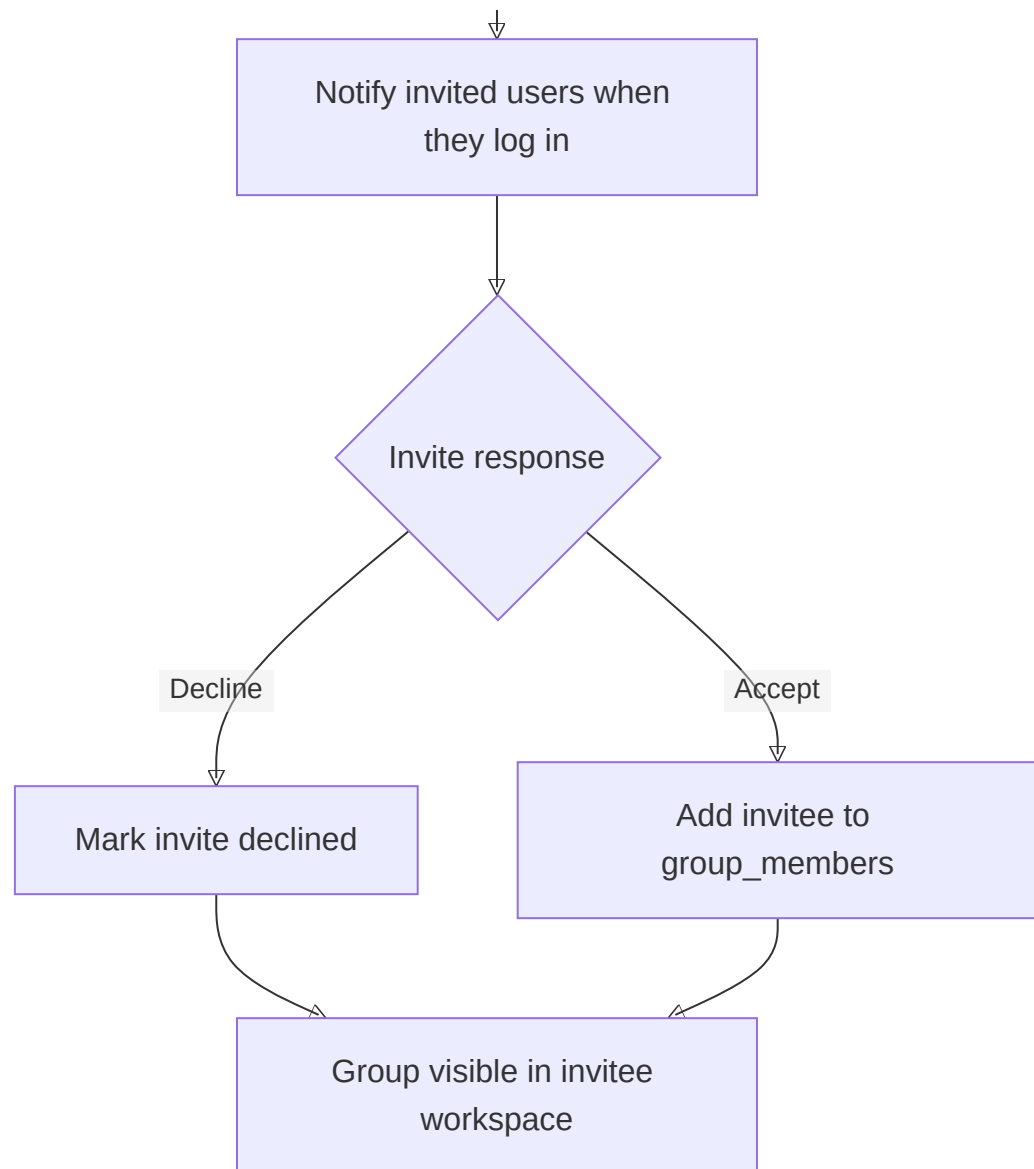
Activity Diagram: Register and Log In



Activity Diagram: Create Group and Invite Members

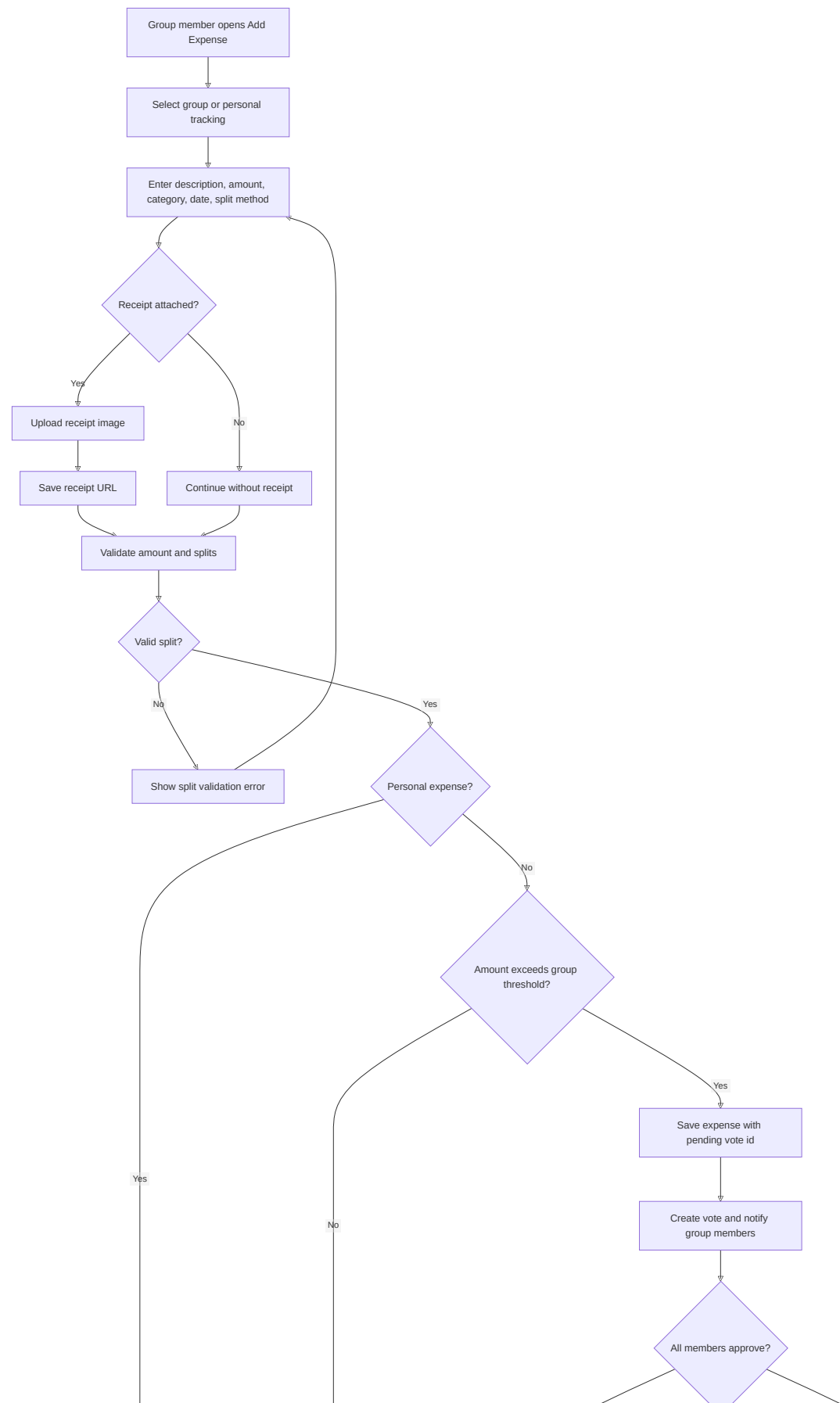


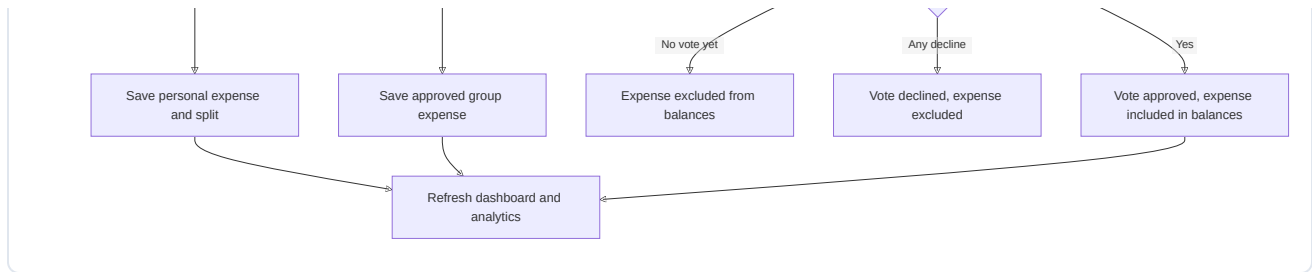




Activity Diagram: Add Expense and Apply Voting Rule

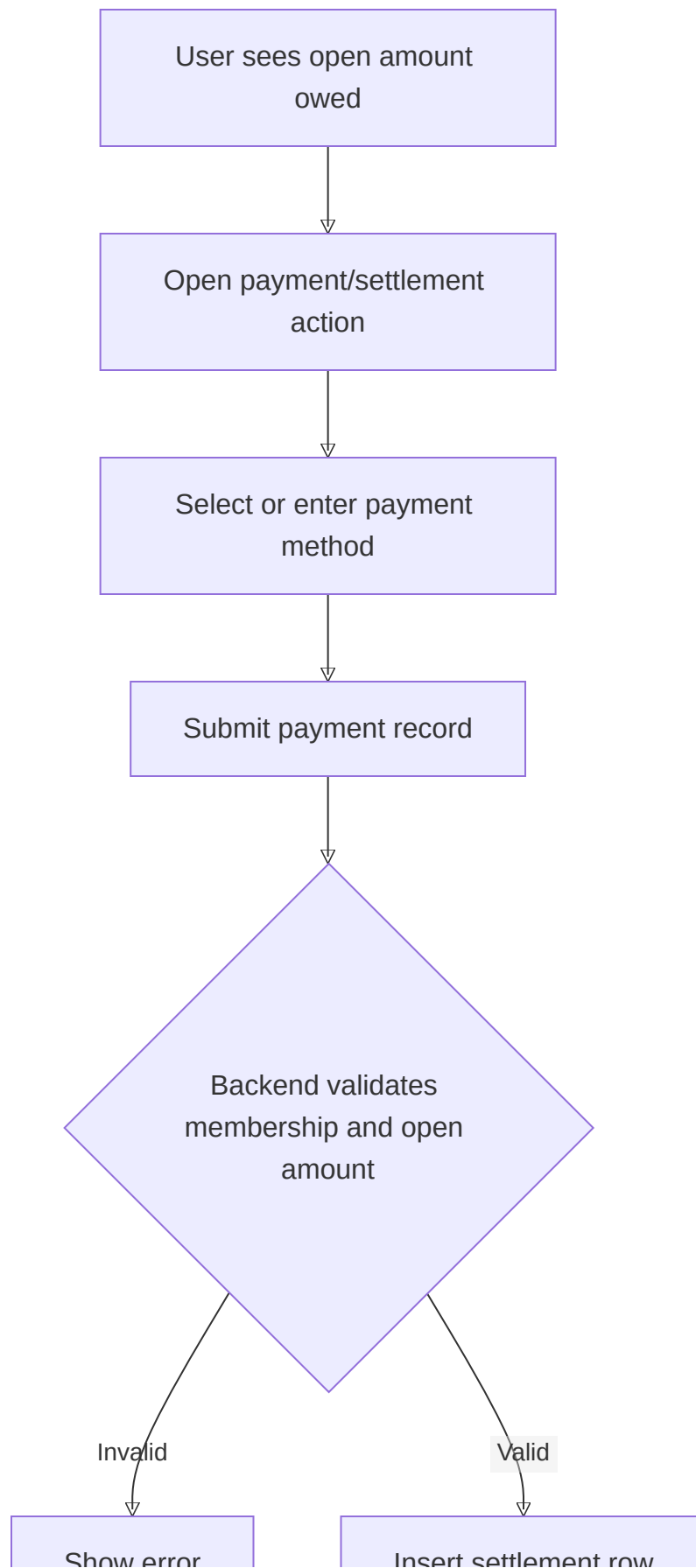


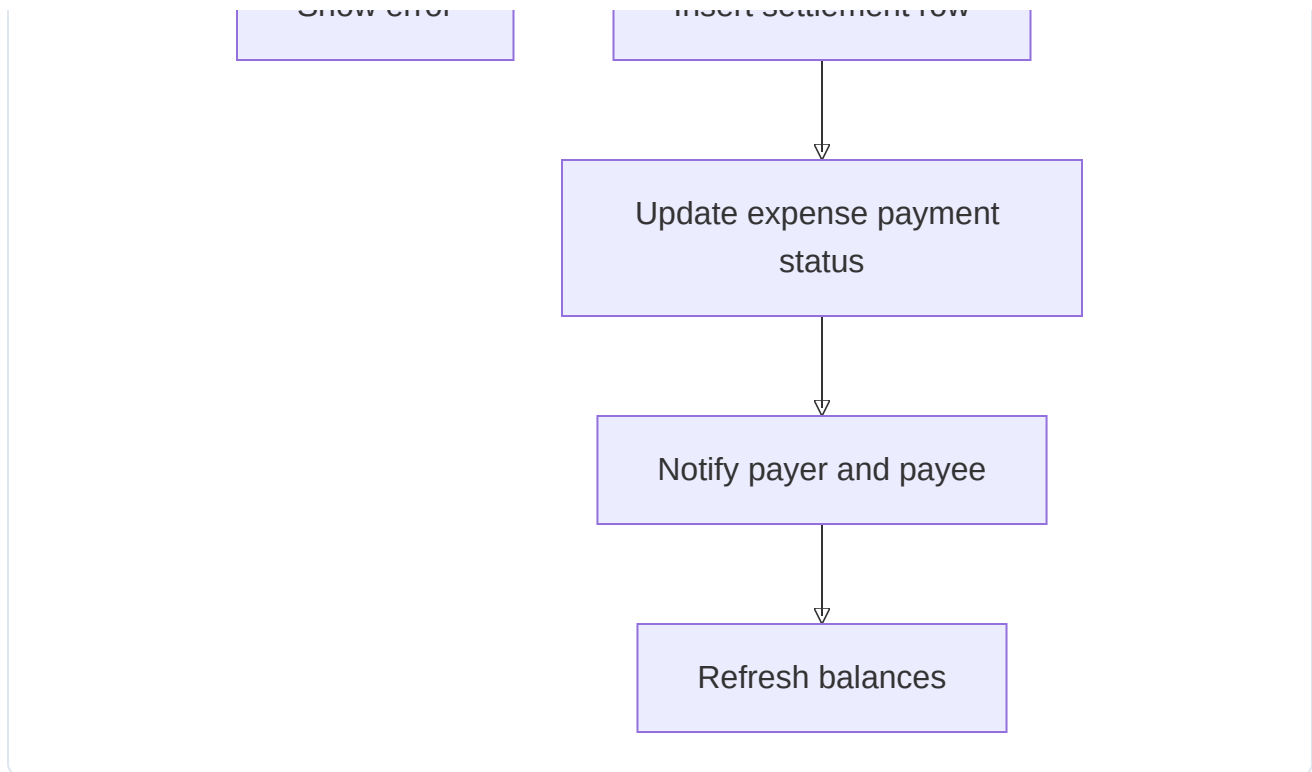




Activity Diagram: Record Settlement

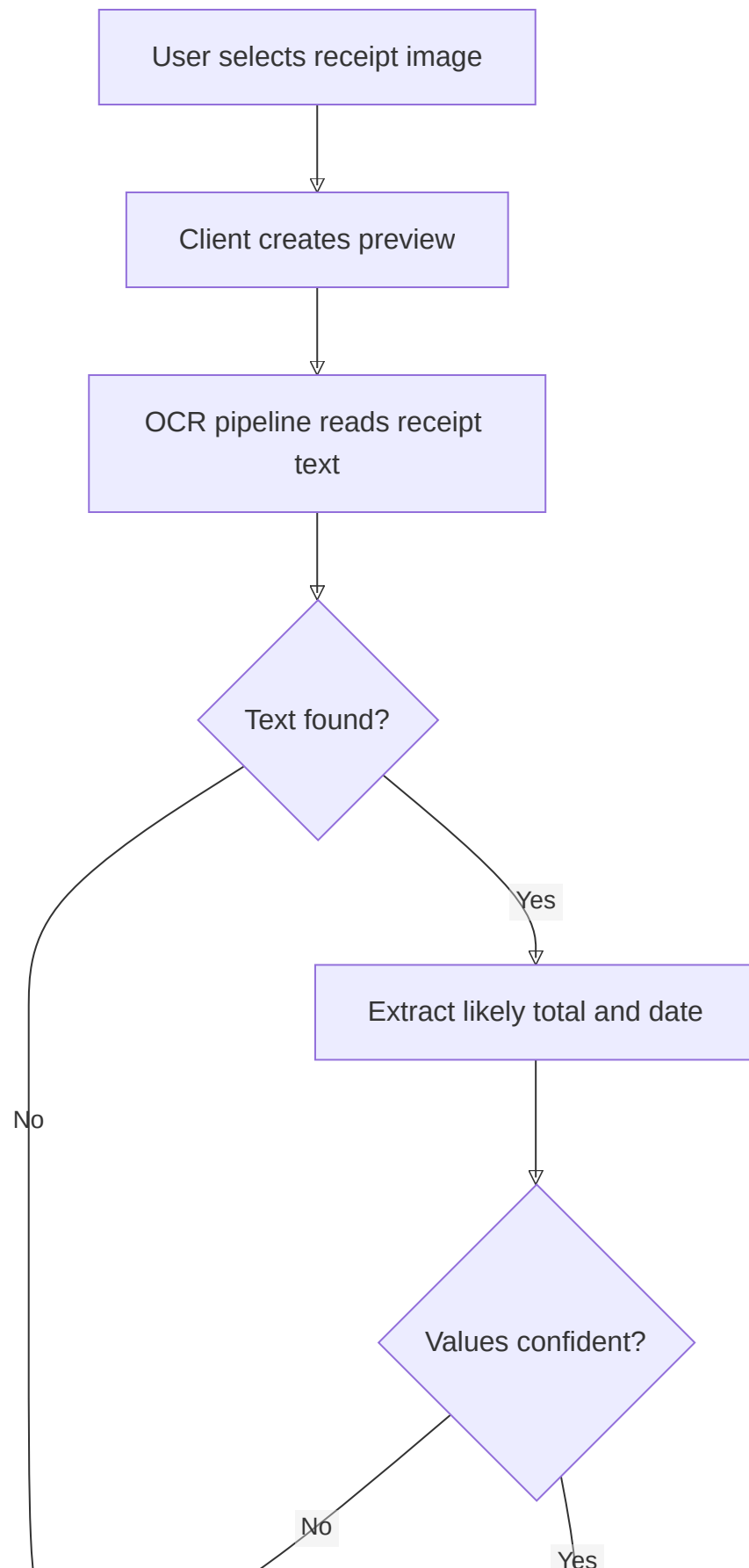


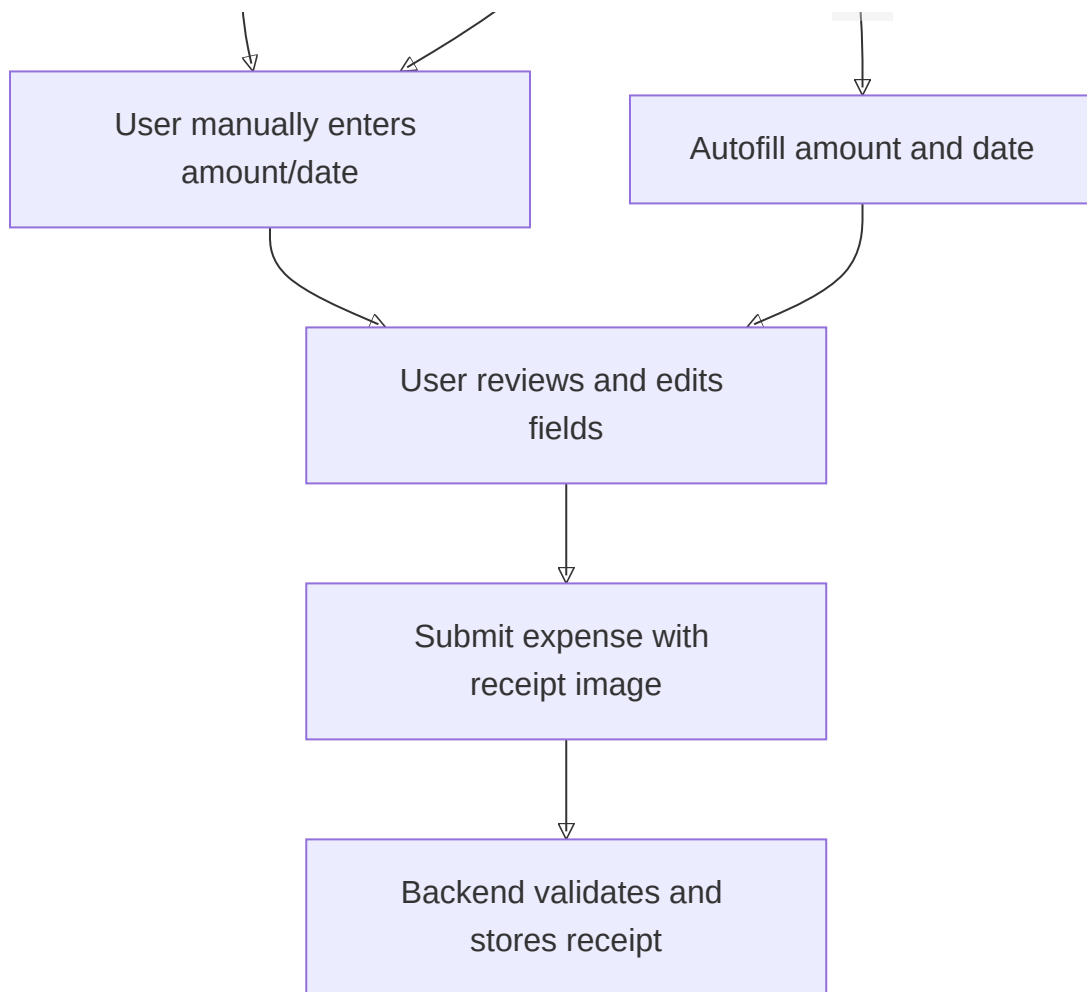




Activity Diagram: Receipt OCR-Assisted Expense Entry

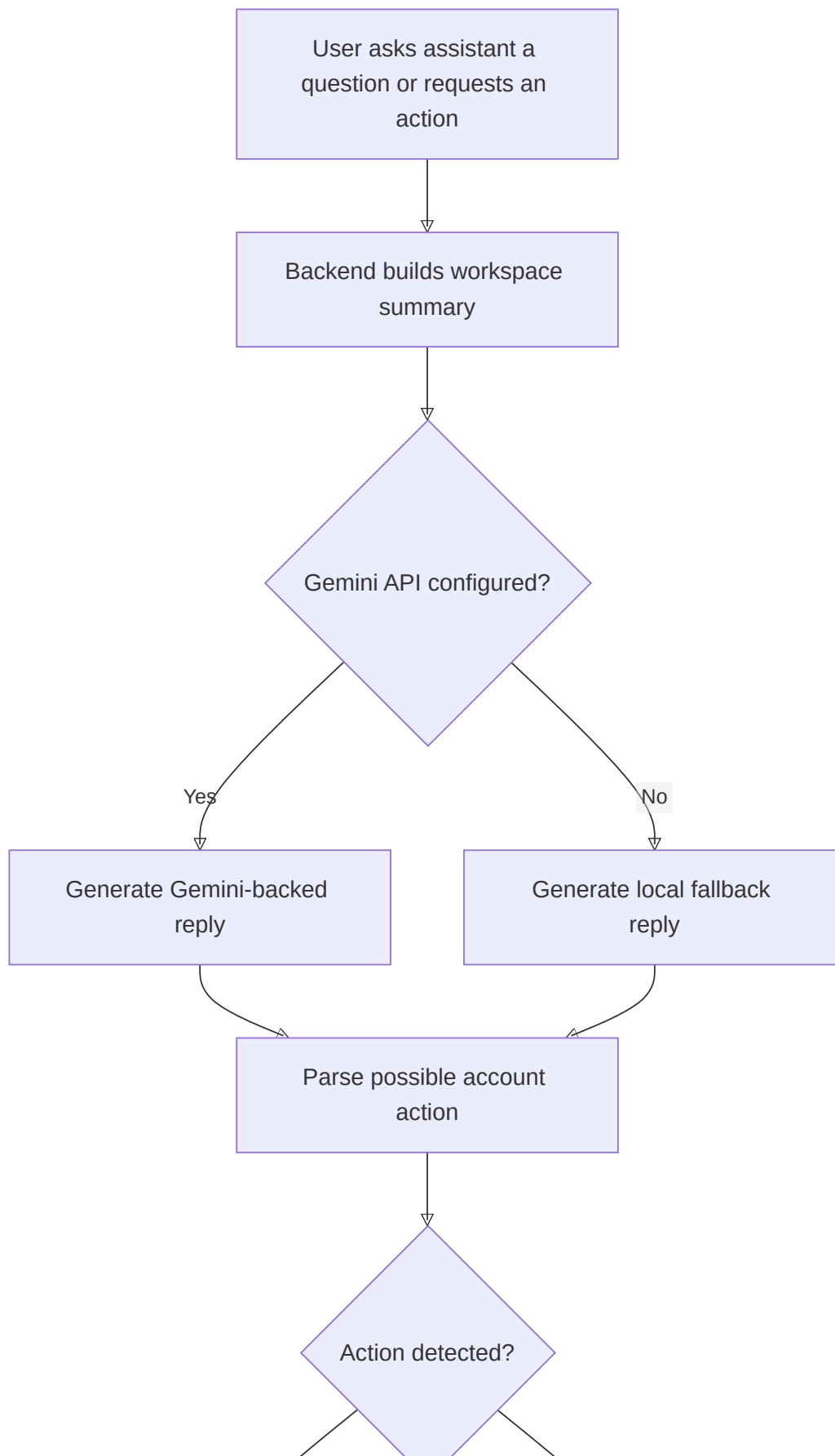


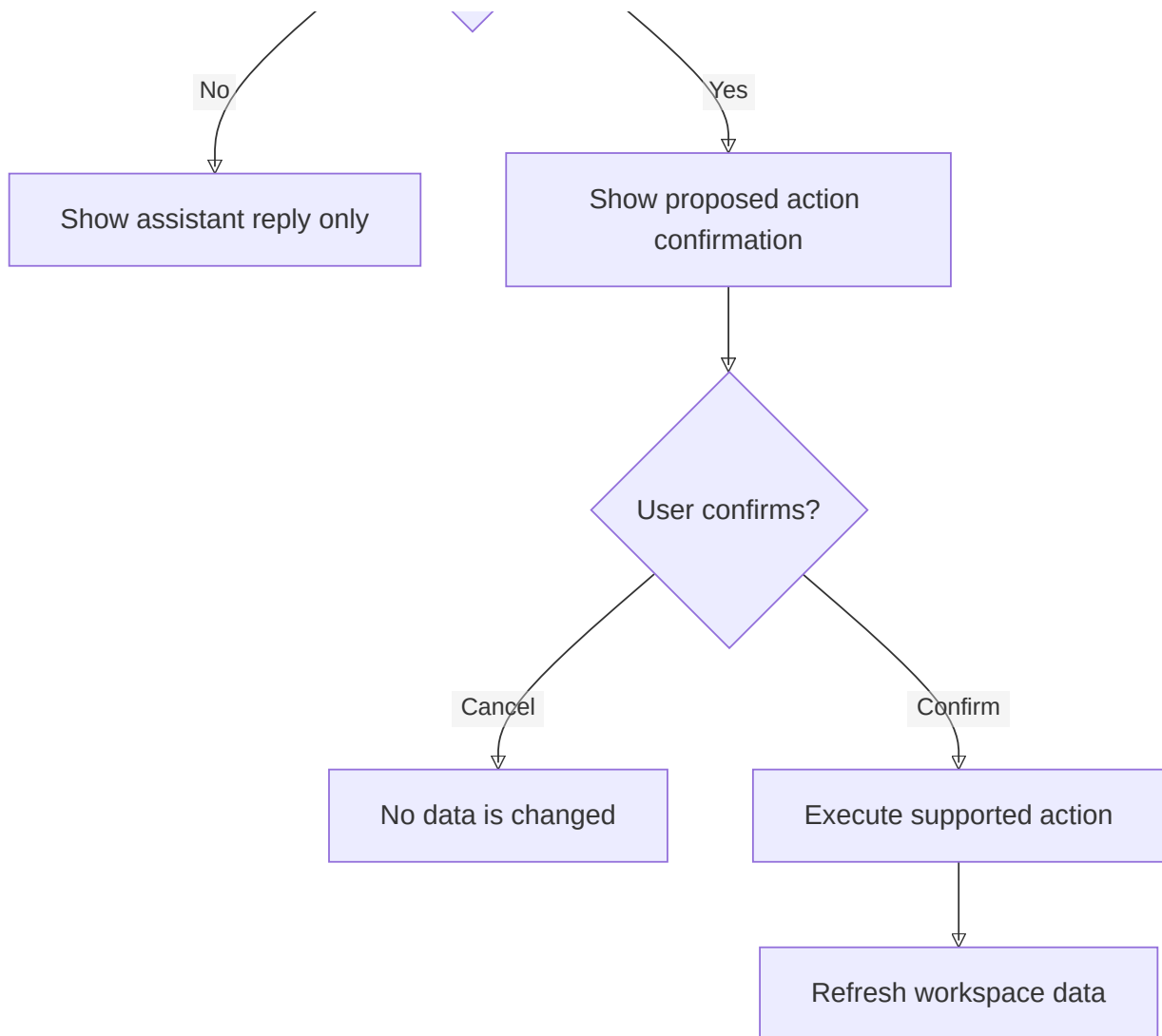




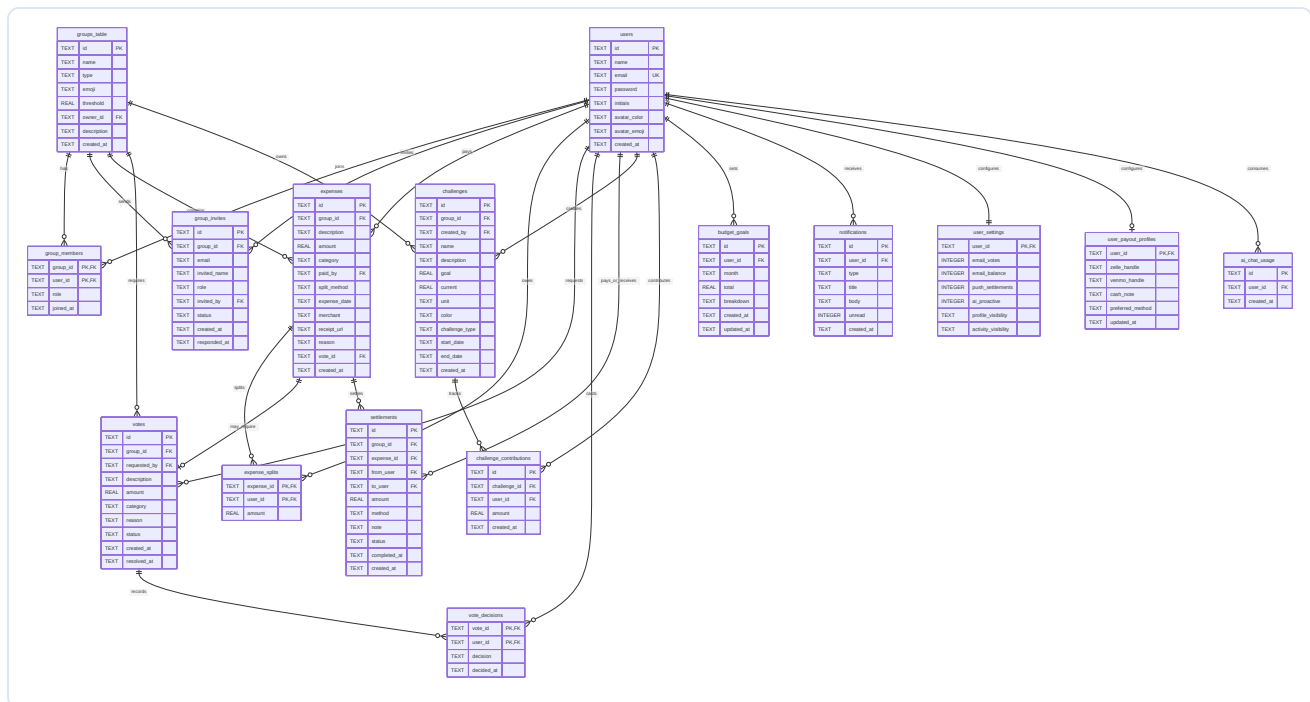
Activity Diagram: AI Assistant Confirmed Action







Domain ER Diagram



Domain Object Summary

Object	Purpose
User	Account identity, profile, avatar, and authentication owner.
Group	Shared workspace with type, threshold, owner, and description.
GroupMember	Membership and role assignment for a user in a group.
GroupInvite	Pending or completed invitation to join a group.
Expense	Recorded purchase, personal cost, or shared group cost.
ExpenseSplit	Per-user amount owed for an expense.
Vote	Approval workflow for above-threshold expenses.
VoteDecision	A member's yes/no response to a vote.
Settlement	Recorded payment against an outstanding expense amount.
Challenge	Group savings/spending goal with progress.
ChallengeContribution	User contribution toward a challenge.
BudgetGoal	User monthly budget and category breakdown.
Notification	In-app event message for invites, votes, balances, and challenges.
UserSettings	Notification and privacy preferences.